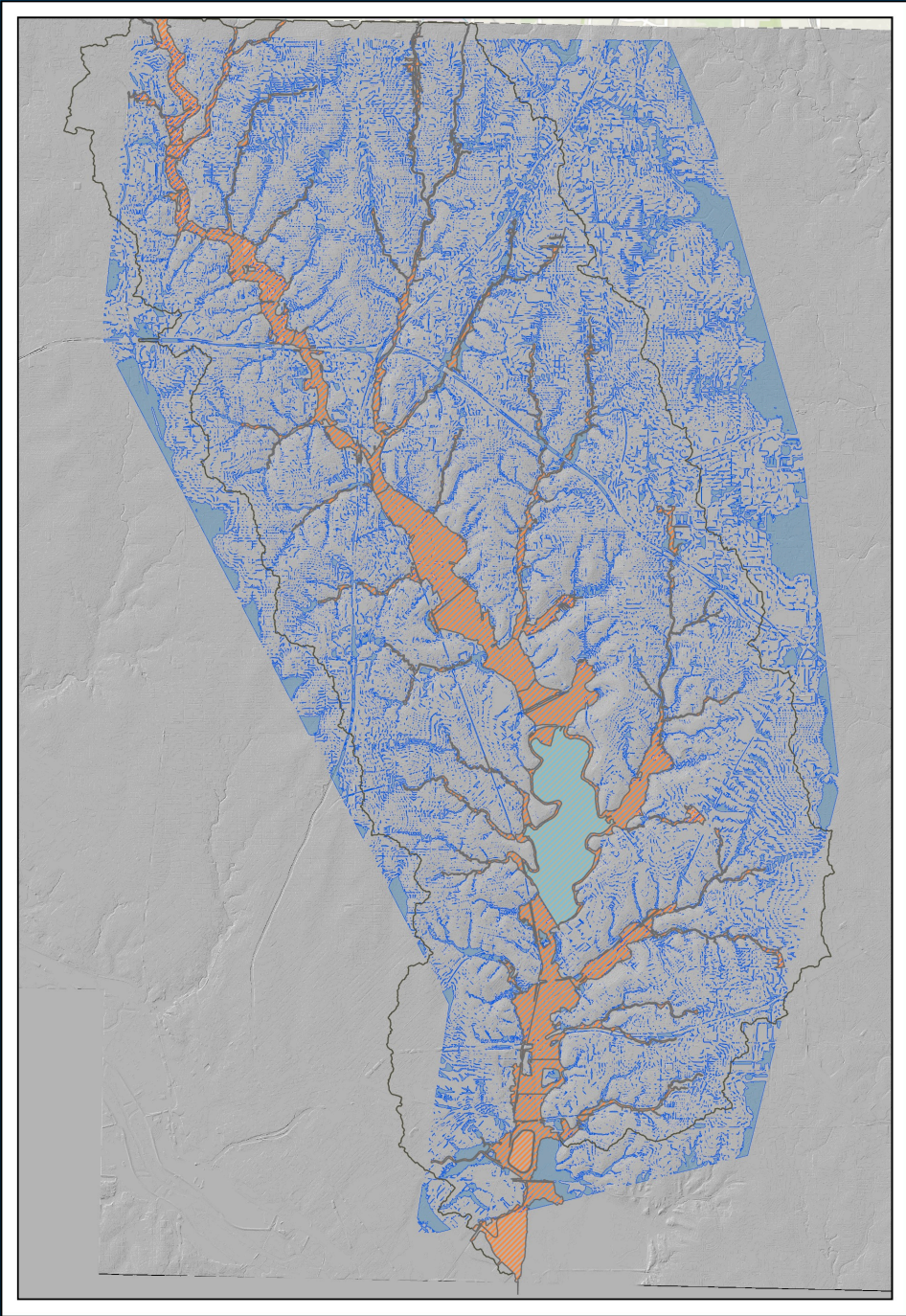


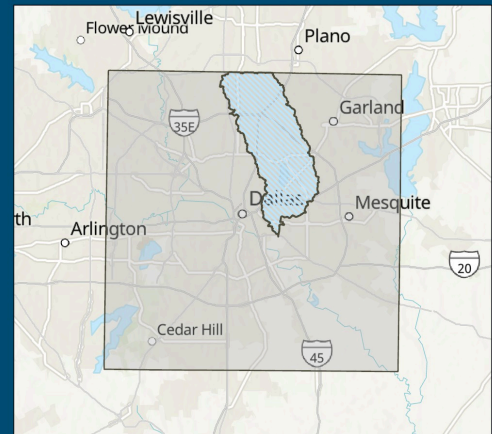
Modeled Floodplain vs FEMA Regulatory Zones

HEC-RAS 2D Rain-on-Grid Model vs FEMA NFHL A, AE, AO Zones (100-Year)



Legend

-  Modeled 100-yr
-  FEMA Zones (A, AE, AO)
-  White Rock Lake Watershed
-  White Rock Lake



Modeled vs FEMA Regulatory Floodplain

100-Year Floodplain	Area (acres)
FEMA 100-yr zones (A/AE/AO)	6,276
HEC-RAS 2D modeled	13,480
Difference	+7,204 (+115%)

Interpretation:

FEMA's regulatory floodplain is almost entirely detailed-study (AE) along main channels. The rain-on-grid model additionally captures pluvial and tributary flooding watershed-wide; the larger extent reflects method scope, not model error.

0 1.75 3.5 7 10.5 14 Miles

Author: Joshua McCulley
Date: 06/20
Projection: NAD83 State Plane TX North Central (FIPS 4202 / WKID 2276) | DEM: USGS 3DEP | Precipitation: NOAA Atlas 14 | Hydraulics: HEC-RAS 7.0.1 | Demographics: US Census ACS 2024 5-yr